

React Basics Notes

Introduction to React

React is a popular JavaScript library used for building user interfaces, especially single-page applications (SPAs). It was developed by Meta Platforms and allows developers to create reusable UI components.

Why React? Component-based architecture Reusable code Fast rendering using Virtual DOM Strong ecosystem and community support Easy state management with hooks **React**

Components

Components are the building blocks of a React application. A component is an independent and reusable piece of UI.

JSX

JSX stands for JavaScript XML. It allows developers to write HTML-like syntax inside JavaScript.

JSX Rules JSX must return a single parent element. Use className instead of class. JavaScript expressions are written inside {}.

Props (Properties) are used to pass data from a parent component to a child component.

React Hooks

Hooks allow functional components to use React features.

Types of Hooks useState: Manages state. useEffect: Handles side effects. useRef: Stores mutable values without re-rendering. useContext: Shares data globally.

Virtual DOM

The Virtual DOM is a lightweight copy of the real DOM.

How It Works State changes. React creates a new Virtual DOM. React compares it with the previous Virtual DOM. Only changed elements are updated in the real DOM. This process is called **Reconciliation**.

Advantages of React Reusable components Better code organization Fast rendering with Virtual DOM Large ecosystem Strong community support Easy integration with APIs Declarative programming model